DA3408 Assignment-01

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1 Conceptual : Technical Debt Diagnosis

1.1 Technical Debt Diagnosis

1.1.a This falls into the Entanglement(CACE) category because altering one feature value is changing some other unrelated feature.

1.1.b This falls into Undeclared Consumers category because some other team is silently consuming the model's output.

1.1.c This falls into Configuration and Glue-Code debt category because there is no single source of truth without any proper structure and so will be difficult to reproduce.

1.2 Proposed Mitigation

For the 1st debt/issue i.e, 'changing the "estimated delivery time" feature's rounding logic unexpectedly hurt recommendations for a completely unrelated "favorite restaurants" feature' , we could use MLflow experiment tracking so that we could catch the unexpected changes by comparing the runs with different parameters.

2 Applied : MLFlow Experiment Comparison

- I conducted six experiments varying the hidden layer sizes and learning rate. The best-performing model utilized a (64,) hidden layer and a 0.001 learning rate. While it did not achieve the highest raw accuracy, it yielded the lowest validation loss, which I prioritized as the primary metric. This is likely due to the smaller architecture's ability to generalize better rather than over-parameterize the problem.
- There was clear evidence of overfitting when analyzing the train_loss vs. val_accuracy trend. Architectures with larger hidden layer sizes achieved higher validation accuracy and lower training loss, but their validation loss spiked. This divergence indicates the wider networks were memorizing the training data instead of generalizing.
- When observing hyperparameter impacts, increasing the hidden_layer_sizes produced a consistent trend of increasing validation loss. Conversely, altering the learning rate yielded mixed results on the loss depending on the architecture. Therefore, based on the primary metric of validation loss, the hidden layer size had the greater, more predictable impact on overall model performance.

Logging Code Snippet: The entire code is in the git repo (see README).

3 Applied: DVC Data Versioning & Rollback

Note: The terminal output and screenshots demonstrating the DVC versioning, data updates, and successful rollback to v1 (with exact row counts) are documented in the GitHub repository (see README).

4 Capstone: End-to-End Reproducibility Drill

I was the partner B, and I ran the experiment my partner setup using the specific commands mentioned in the question and have obtained the same accuracy as his run (upto 4 decimal places).